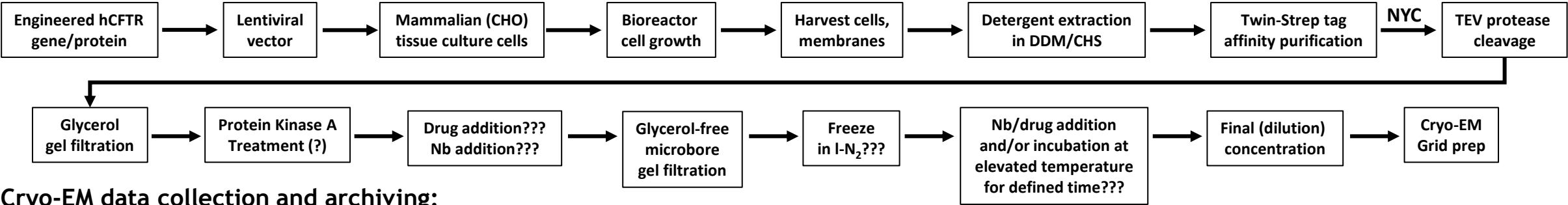
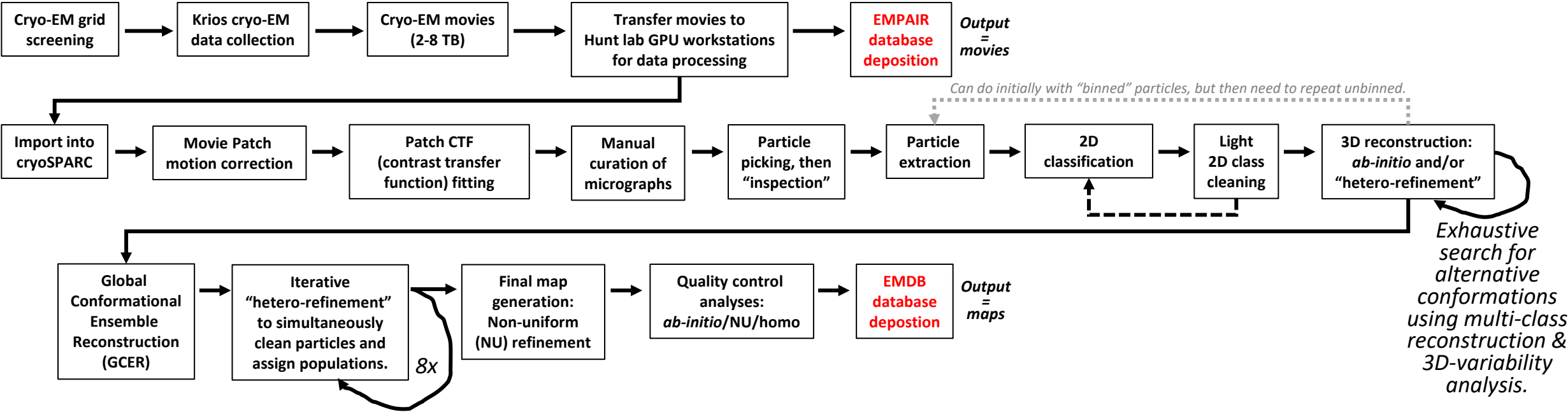


Cryo-EM data processing, global conformational ensemble reconstruction (GCER), and cryo-EM map archiving in cryoSPARC:

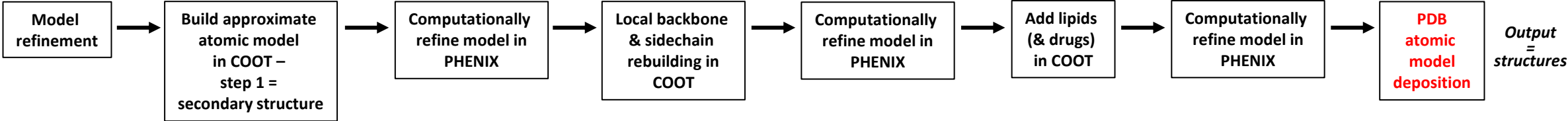
University of Alabama Birmingham / Columbia hCFTR protein production and purification pipeline:



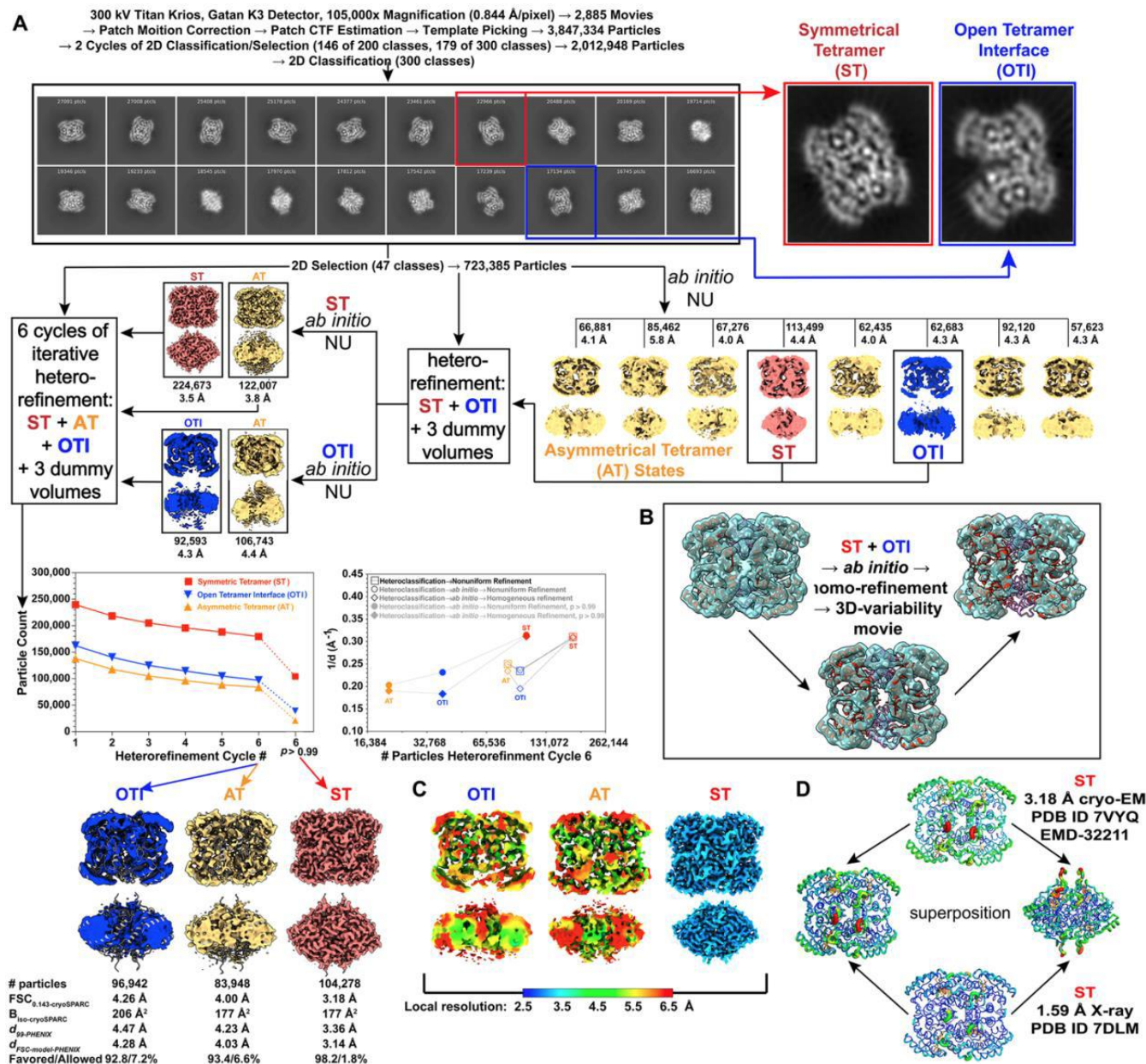
Cryo-EM data collection and archiving:



Cryo-EM atomic coordinate model building & refinement in COOT/PHENIX and archiving:



Example classification/structure figure from Li *et al.* (2022) EMBO J. 41:e108368 (<https://doi.org/10.15252/emboj.2021108368>):



Standard local resolution map coloring scheme shown here:

- blue = 2.5
- cyan = 3.5
- green = 4.5
- yellow = 5.5
- red = 6.5

In ChimeraX:

- “Volume” -> “Surface Color”
- On “Color surface” line, pick map and by “volume data value”
- On next line, select locres map for “using map”
- Activate “Options”, and select 5 colors
- Manually input colors/values above
- Hit “Color” button
- Hit “Key” button
- Go to Graphics menu
- Select white background, full lighting, no shadows
- Go to Home menu, and save image in PNG format

hCFTR conformational state nomenclature:

Symmetrical Closed (SC) = Symmetrical Occluded = Narrow 0° (N0) – symmetrical NBDs, narrow TMDs with internally formed chloride channel that is occluded at the extracellular surface
Asymmetrical Open (SO) = Asymmetrical NBDs, narrow TMDs with TM8 ostensibly disordered in the extracellular leaflet of the membrane

Representative reference maps for Heterorefinement and corresponding models:

SC map: <https://www.dropbox.com/s/cvez6yevxn2s5au/D1247lppATPN0.mrc?dl=0>

SC model: https://www.dropbox.com/s/copty9bbcbvvin2/D1247lppATPN0_Prsr047.pdb?dl=0

AO map: https://www.dropbox.com/s/0xkmbm8dcswwcht/cryosparc_P14_J1305_005_volume_map_sharp_local_AO_zflipped.mrc?dl=0

AO model: https://www.dropbox.com/s/v6rvh5w7cnh978c/D1247_AO_to_P14J1305-coot-5c.pdb?dl=0

NBD1less map: <https://www.dropbox.com/s/g0napg6oqkijzm0/D1177ATPVX2PU.mrc?dl=0>

NBD1less model: https://www.dropbox.com/s/ldtq740t6q8dgtw/D1177ATPVX2PU_Prsr033b_Prsr072_Prsr080.pdb?dl=0

V21 map: <https://www.dropbox.com/s/ii4b0i19f59i96o/D1177ATPVX2V21.mrc?dl=0>

V21 model: https://www.dropbox.com/s/mz4hhwm47d052c5/D1177ATPVX2V21_Prsr038c_Prsr070-coot-2_real_space_refined_070.pdb?dl=0

V21-LipidRaft map: <https://www.dropbox.com/s/anp8uq361m3fc3e/D1177ATPVX1V21LR.mrc?dl=0>

V21-LipidRaft model: https://www.dropbox.com/s/yn9704julo53t4k/D1177ATPVX1V21LR_Prsr054_Prsr007_Prsr078.pdb?dl=0

Standard dummy maps (volumes) for Heterorefinement:

#0: https://www.dropbox.com/s/unvvmkvek3bqmh0/DecoyEmptyMicelle_P14_J93_modif.mrc?dl=0

(empty micelle)

#1: https://www.dropbox.com/s/kjyghdrrovqqqlw/1DecoyGPCR_P11_J28_modif.mrc?dl=0

(low resolution GPCR complex)

#2: https://www.dropbox.com/s/6vkmt8fedkq673q/2DecoyAbInitio1_P14_J1076_class_01_final_volume.mrc?dl=0

(smaller origin-shifted solid particle)

#3: https://www.dropbox.com/s/ge54hwj8yy0gqnm/3Cryosparc_P10_J1222_class_01_final_volume.mrc?dl=0

(low resolution transmembrane domain in a micelle)

#4: https://www.dropbox.com/s/e871rw6gpwo11vl/4Cryosparc_P10_J1226_class_00_final_volume.mrc?dl=0

(low resolution transmembrane domain in a micelle)

#5: https://www.dropbox.com/s/vt5mv597g43tzzf/5Cryosparc_P10_J1227_class_01_final_volume.mrc?dl=0

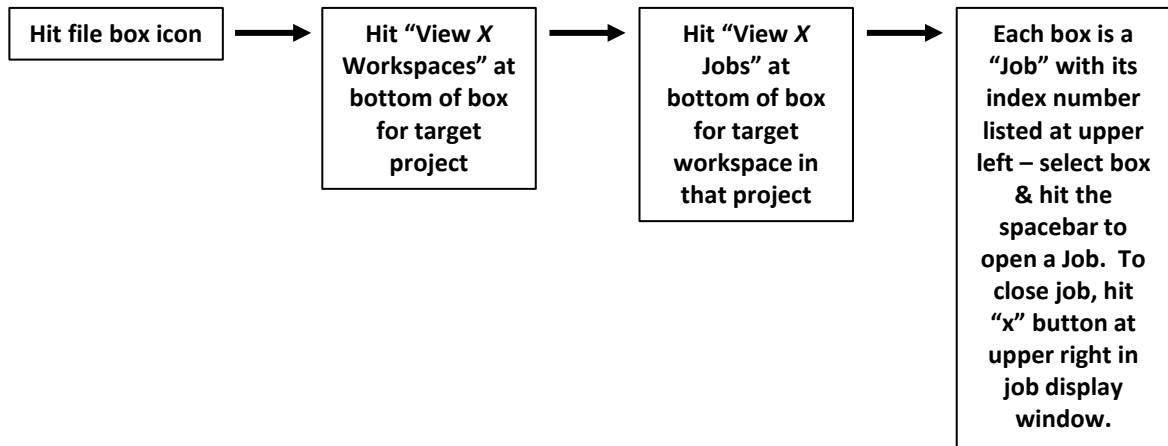
(low resolution transmembrane domain in a micelle)

hCFTR subdomain boundaries:

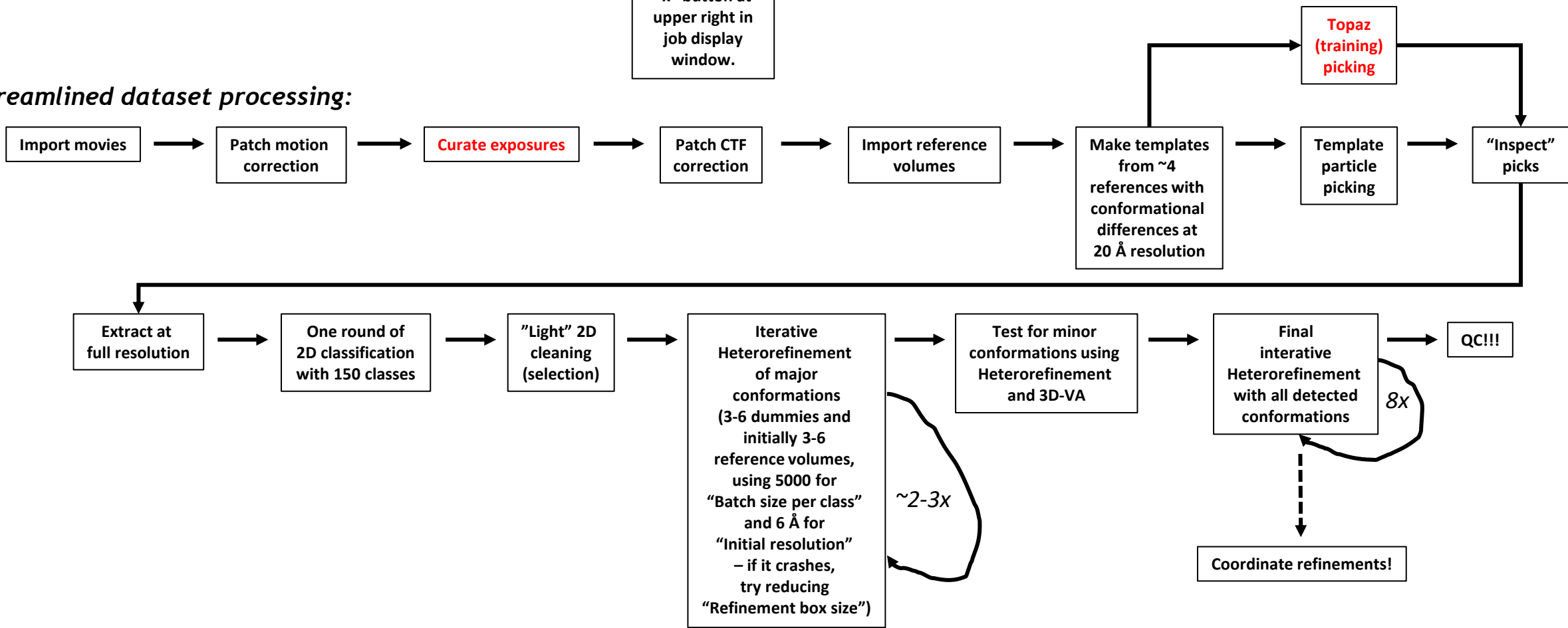
LassoNter = 1-45, LassoCter = 46-77, TMD1= 78-218 & 333-380 & 1013-1120, CORE1 = 391-494 & 566-638, (RI = 405-436), AHD1= 495-565, WalkerB1 = 578-596,
R-Region: 639-845, TMD2 = 221-328 & 846-1110 & 1129-1173, 639-CORE2 = 1202-1292 & 1364-1427, AHD2=1293-1363, WalkerB2 = 1377-1393

Standard cryoSPARC workflows:

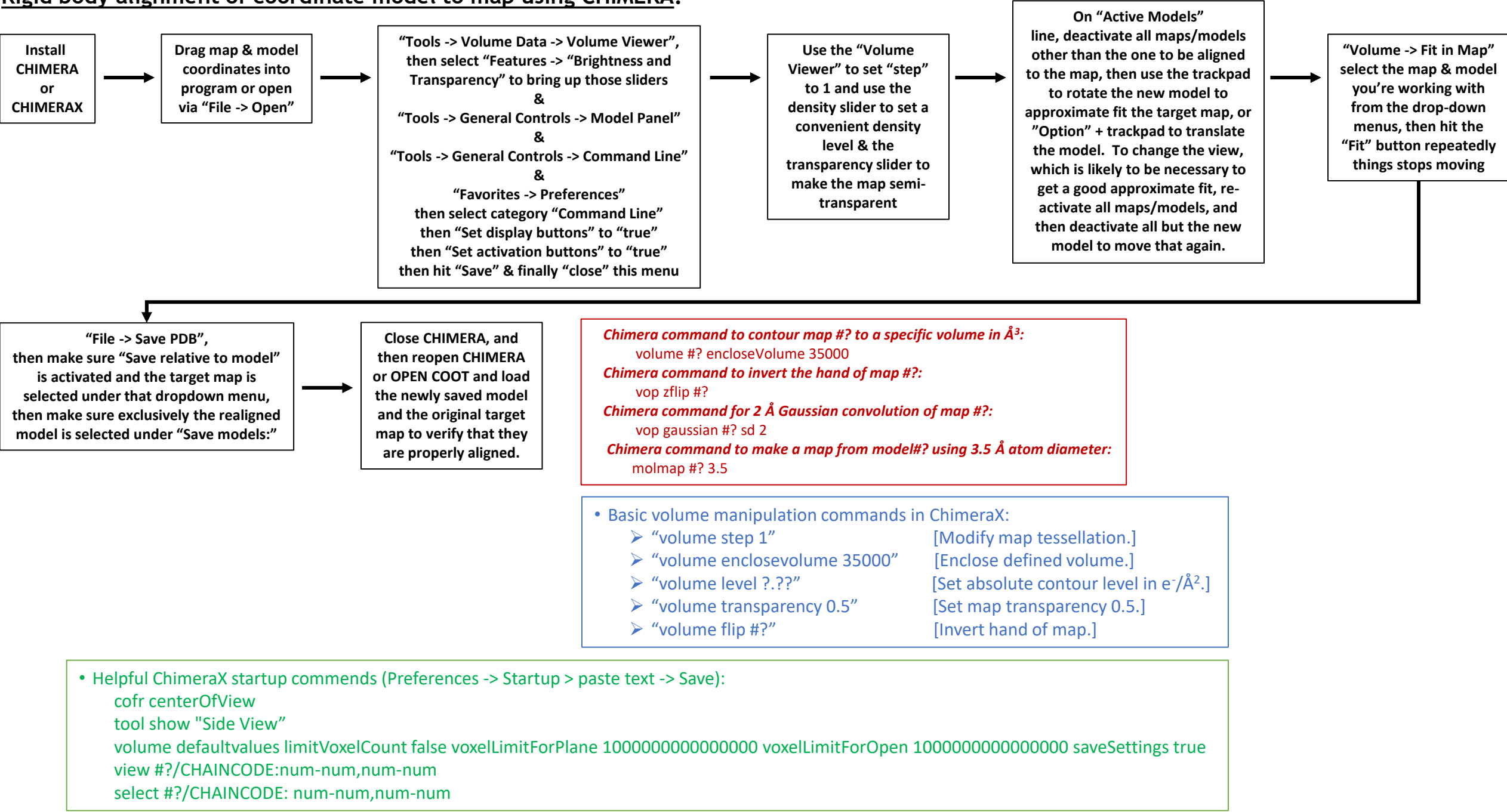
View Project/Workspaces/Jobs:



Streamlined dataset processing:



Rigid body alignment of coordinate model to map using CHIMERA:



Run movies in ChimeraX:

- “open PATH/ROOTFILENAME*.mrc vseries true”
- “volume step 1”
- “volume enclosevolume 35000”
- “vseries play #? loop true direction oscillate pauseframes 10 normalize true”
- “vseries stop #1”
- “vseries slider #1”

<u>ChimeraX zflip command:</u> volume flip #?
--

Delete bond in ChimeraX:

- ~bond #21/V:1@O33/A:68@CE

Color cryo-EM density in ChimeraX:

Colors must be assigned via a model aligned to the map (named chain in the example below)

- color /A:1-67 cyan atoms
- color /A:68-383 cornflower atoms
- color /A:384-650 0,0,60 atoms
- color /A:840-1173 lime atoms
- color /A:1202-1460 dark green atoms
- color /A:501-566 0,20,100 atoms
- color /A:1299-1364 sea green atoms

#Tools>Volume Data>Color Zone

chimeraX color names: <http://rbvi.ucsf.edu/chimerax/docs/user/commands/colornames.html>

Enable OccuPy to launch ChimeraX:

- Run the following command once to add the ChimeraX executable location to your PATH:
`echo 'export PATH=$PATH:~/Applications/ChimeraX-1.7.1.app/Contents/bin/' >> ~/.zshenv`

Evaluate/fix pixel size:

- Get best structure available covering at least 100 residues, ideally a high-resolution crystal structure. Align (fit) that model to the density in ChimeraX.
- Activate “Fit in Map” and “Map Coordinates” menus.
- In “Fit in Map” menu, select “Real-time correlation”, “Fit map simulated from atoms” (~3 Å), & “Use only data above contour level”, and check “correlation” on “Optimize line”.
- In the “Map Coordinates” menu, vary the pixel size in 0.01 Å steps, running the “Fit” command in the “Fit in Map” menu each time.
- Record the correlation value at each pixel size, and find the pixel size that gives the maximum correlation.
- If different from the pixel size listed in the input map, save the map in ChimeraX after changing the pixel size to the optimal value using the “Map Coordinates” menu.
- Fit the current model to that map after re-uploading it in ChimeraX, and save that realigned model to disk relative to the map with the updated pixel size.
- Refine that map and model against each other in PHENIX.

Import & preprocess movies in cryoSPARC:

- Get the best estimate possible of the pixel size – ideally from a recent high-resolution-coordinated-based refinement of a map from recently collected dataset.
- Import Movies using that pixel size and the total electron dose (based on the exposure protocol), ideally also including the beam-shift records from xml files.
- Run Patch Motion Correction plotting 50 movies and using at least 100 movies to train the denoiser model, running on 4 GPUs if possible.
- Run Patch CTF Correction plotting 50, running on 4 GPUs if possible.
- Run Micrograph Denoiser using at least 100 movies for training and as many CPUs as available (*e.g.*, 48 if possible).
- Run template picking if good reference volumes are available, using default parameters for template generation (with a 200 Å particle diameter for hCFTR).

Moving cryoSPARC particles and volumes between workstations:

- Select/highlight the job in the cS Workspace, and then go to "Actions" button at bottom right and select "export".
- The data from each job will be written to a directory starting with job # located within the exports/jobs subdirectories within the cS directory for the source Project.
- When exporting particles from an extraction job, you should remove the links to the corresponding movies before exporting to avoid also copying the movies:
 - Run a Particle Sets job using the low-level inputs menu to remove the "locations" data from the input, which will disconnect the movies.
 - That job will "split" the input target particle set into a single batch without any links to the original movies.
- All subdirectories under the exports/jobs subdirectory for the source project must be copied to the import/jobs subdirectory for the target project using one of several possible methods:
 - Create and a tar archive including all of the exported jobs in the exports/jobs subdirectory for the source project, and move it to the import/jobs subdirectory for the target project:
 - Other than for NU jobs, preserve all soft-links when creating the tar archive by using the `-h` flag, *e.g.*, `cd` to exports/jobs and execute `"tar -cvhf ../cSExportedData.tar."`).
 - Compressing the resulting tar archive file using `gzip` will accelerate transfer if it needs to be moved to another workstation or NAS.
 - If the jobs are being transferred to workstation that cannot access the disk on which the tar archive file was created, copy it to a disk visible on the target workstation.
 - If needed, unzip the tar file.
 - Then extract the tar file to the imports/jobs subdirectory for the target project (*e.g.*, `cd` to imports/jobs and run `"tar -xvf filename.tar"` or use the NAS GUI).
 - Alternatively, use `rsync` with flags to preserve soft-links to copy the source project exports/jobs directory to the target project imports/jobs directory:
 - *E.g.*, `"rsync -avh --info=progress2 exports/jobs uid@workstation.biology.columbia.edu:/cS-directory-path/imports/jobs",` followed by password authentication.
- Go to the cryoSPARC Project/Workspace into which you want to import the job, and select "Import Job" under the green "New Jobs" button near the upper right.
 - That action will bring up a menu with a file navigator that can be used to find the directory containing the job (which will begin its J#) in the imports/job directory.

Connecting beam-shift groups to cryoSPARC exposure sets:

- Using a compressed EPU-like xml directory:
 - Copy the pathname for the cryoSPARC Project Directory by clicking on that field on the right side of the cryoSPARC interface.
 - Move the compressed xml directory to the uploads folder in the directory tree for the cryoSPARC project:
 - One option is to drag-and-drop the compressed xml directory onto the cryoSPARC interface and then upload it from there.
 - Alternatively, sftp to a workstation on which the disk containing the cryoSPARC Project Directory is mounted, then:
 - ❑ local_computer> cd local_directory_containing_compressed_xml_directory
 - ❑ local_computer> sftp uid@workstation.biology.columbia.edu
 - ❑ remote_computer> cd cryoSPARC_project_dir/uploads
 - ❑ remote_computer> put compressed_xml_directory.zip
 - Decompress that the compressed xml directory:
 - Use ssh to log in to a workstation on which the disk containing the cryoSPARC Project Directory is mounted, then:
 - ❑ ssh uid@workstation.biology.columbia.edu
 - ❑ remote_computer> cd cryoSPARC_project_dir/uploads
 - ❑ remote_computer> unzip compressed_xml_directory.zip
 - Run an Import Beam Shift job in cryoSPARC:
 - Set the pathname for the job to cryoSPARC_project_dir/uploads/*.xml to capture all of the xml files in the directory.
 - Trailing characters and potentially also leading characters have to be stripped from the movie and xml filenames to get them to match exactly.
 - This is typically 11 trailing characters (.frames.tif) for the movie files, 4 trailing characters (.xml) for the xml files, and no leading characters for either.
 - Run an Exposure Group Utilities job in cryoSPARC to cluster the beam-shift values into groups:
 - Use kmeans clustering, typically with 36-100 groups.
 - Input Selection = exposure.
 - Action = cluster&split.
 - Combine Strategy = take mode.
 - Correspond particles to exposures and enforce consistency of exposure group IDs = true.
- Using an old-style Leginon rtf file containing one set of exposure and beam-shift values per line:
 - First step is to convert the rtf file to an xml directory:
 - Use TextEdit or another rtf file editor to remove any header or tailing lines from the rtf file so that it contains exclusively lines with movie names and two beam coordinates.
 - Convert it to a plain text file = “Format” -> “Make Plain Text” in TextEdit, and save it as plain text file.
 - Covert it to csv format using awk and then check the contents using more:
 - ❑ localhost> '{printf "%s, %s, %s\n", \$2, \$3, \$4}' leginon_filename.rtf > leginon_filename.csv
 - Copy Kook's Python script ([csv2xml.py](#)) to the working directory, use it to generate an xml directory from, and check that the number of xml files matches the number of movies:
 - ❑ localhost> ./csv2xml.py leginon_filename.csv
 - Compress the resulting directory, *e.g.*, by right-clicking on the directory in the file browser on a Mac or PC.
 - Follow instructions above for using a compressed EPU-like xml directory.

Reconnecting beam-shift groups to cryoSPARC exposure sets when they are lost (*e.g.*, after RBMC):

- Rerun Exposure Group Utilities using particles from all conformations to be refined and the most output of recent exposures with beam shift values (*e.g.*, all from RBMC).
- Run Particle Sets Tools to get the “intersection” between those output particles and those from an individual conformational to be refined.
- Use the particles in the output for that job in the intersection set belonging to the Exposure Group Utilities output for further processing.
- Run NU using those particles together with the most recent volume for the individual conformational class to be refined and turning on “Grouped CTF refinement”.

MASK creation for cryoSPARC 3D-VA, 3D classification, & local refinement:

- Move target map and particles into reference orientation:
- To move target map into reference orientation, download (map_sharp) and then load reference and target maps into ChimeraX on local computer:
 - Align target map to reference map by first selecting target map and using the “right mouse” to approximately align it to the reference.
 - Optimize alignment of the maps using the “Fit” icon under the map menu.
 - Resample target map on reference map grid – “volume resample #t ongrid #r” with #t being target map number and #r being reference map number.
- Load saved map back into ChimeraX on local computer to check that it’s correct.
- Move the target particles into the reference orientation
 - Use sftp to transfer the reoriented target map from your local computer to the NAS. (See sftp instructions below.)
 - Import the reoriented target map from the NAS into cryoSPARC (“Import Volume” job), and then run NU in cryoSPARC.
- Download reoriented target map from NU into ChimeraX on the local computer to verify orientation is correct.
- Align model coordinates to reoriented target map in ChimeraX on the local computer. (Use PyMOL or text editor to limit coordinates to desired envelope region.)
- Make atom map from model:
 - “molmap #p 3.5 ongrid #tr” with #p being pdb model # and #tr being the number of the target map aligned to the reference map.
 - Save molmap output map to disk on local computer, and reload it into ChimeraX to confirm it’s the right orientation.
- Import molmap output into cryoSPARC. (Use sftp to transfer it from your local computer to any LINUX workstation in the lab, and then move it to the NAS).
 - Go (cd) to directory on your local computer with map – “cd PATH”, where PATH is the specification for the directory containing the molmap.
 - Use sftp to connect your local computer to one of the LINUX workstations in the lab – e.g., “sftp uid@west.biology.Columbia.edu”.
 - Upload the molmap output to that LINUX workstation – “put FILENAME”.
 - Check upload looks correct using “ls” within sftp, and then “exit” from sftp on your local computer.
- Move molmap from your home area on that LINUX workstation to the NAS:
 - Log in to that workstation via ssh, e.g., “ssh west.biology.Columbia.edu”.
 - Move the transferred molmap to the NAS – e.g., “mv FILENAME /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/hCFTRReferenceMaps/D1557WThCFTR/”.
- Verify molmap is in the destination directory on the NAS via its web interface – <http://128.59.63.191:5000/> (jfh21 account).
- Load molmap into cryosparc using Import Volumes.
- Convert molmap to mask in cryoSPARC using “Vol Tools” with input=map, output=mask, threshold=0.001, dilation=2, and padding=12 (or padding=15).
- Download mask and use ChimeraX to verify that it is properly aligned to the reoriented target map.
- Run 3D variability analysis in cryoSPARC using “3D Var” with the imported mask and the realigned NU output – modes=5, filter-res=6.
- Generate 3D variability movies in cryoSPARC using “3D Var Display” on output from 3D Var with res=6, output-mode=intermediates, number=7.
- Examine 3D variability movies using Volume Series in ChimeraX:
 - “open PATH/ROOTFILENAME*.mrc vseries true”
 - “volume step 1”
 - “volume enclosevolume 35000”
 - “vseries play #? loop true direction oscillate pauseframes 10 normalize true”
 - “vseries stop #1”
 - “vseries slider #1”

ChimeraX zflip command: volume flip #?

Representative local refinement protocol for hCFTR (*AltNBD1* from *D1557+ATP+T2a* dataset, April 2025):

1. Run NU $\pm$ 8 extra final passes using particle set from heterorefinement at the same stage in the previous cycle:
 - Use particles from iterative heterorefinement, ideally after CTF refinement using beam-tilt groups.
 - If running on an *ardecon* map, best to clone the NU at the same stage in the previous cycle just changing in the input map and running with default or 8 extra final cycles.
 - Map can come either from heterorefinement (default initial resolution) or from *ardecon*, in which case use 4 Å initial resolution (3 NUs total at this stage for *ardecon*).
 - If map is from *ardecon*, generate NUs seeded with maps with no sharpening, PHENIX non-model sharpening, and PHENIX model-based sharpening (because histograms are very different).
 - Compare the maps using the different reference volumes $\pm$ 8 extra final passes to identify which map is best in each region: the TMD, mystery density region, CORE1, AHD1, CORE2, AHD2, & T2a.
2. Run whole domain local refinements:
 - *E.g.*, NBD1+T2a, T2a, & NBD2 in the case of AltNBD1.
 - Use particles from the NU showing the best density in each specific target region.
 - Run local refinements using the reference volumes from the same NU as the particles. If using *ardecon*, also local refinements in parallel using the reference map used to generate each NU.
 - So exactly two parallel locals at this stage if using *ardecon* output (**exactly 8 locals for AltNBD1**), but only a single local if directly using heterorefinement output.
 - Mask parameters = 0.01 threshold, 2 Å dilation, & 12 pixel padding.
 - Default local refinement parameters = 3.5 Å initial resolution, 3 extra final passes, no dynamic masking.
 - Rotation/translation restraints = 3°/2 Å for NBD1+T2a, 4°/4 Å for NBD1 or NBD2 alone.
 - Pick best domain locals to use in subdomain locals.
3. Run subdomain local refinements:
 - *E.g.*, CORE1, AHD1, T2a, CORE2, AHD2, WalkerBHelix1, and possibly WaklerBHelix2 and mystery-density.
 - Run subdomain locals using the particles and reference volume from the domain local showing the best density in the target region. Small difference in reference map not important here.
 - Run two subdomain locals in parallel using the particles and map from both the domain local including that region and the overall NU having the best density in that region.
 - So exactly two parallel locals at this stage (**12 or 16 locals total in the case of AltNBD1**).
 - Mask parameters = 0.01 threshold, 2 Å dilation, & 12 pixel padding for larger subdomains (COREs, and T2a) but 5 pixel padding for smaller subdomains (AHDs, WalkerBHelices, mystery density).
 - Default parameters = 4 Å initial resolution, 3 extra final passes, no dynamic masking.
 - Rotation/translation restraints = 5°/4 Å for CORE1, 4°/4 for AHD1, 3°/4 for T2a, 2°/2 for CORE2, 2°/2 for AHD2, ?°/? for WalkerBHelix 1, ?°/? for WalkerB Helix2, & ?°/? for mystery density.
 - Pick best domain locals to use for composite map synthesis.
4. Make a manual composite map.
 - Use NU with best TMD density to serve as the consensus maps. It's most efficient to use a single local for each region, although it's possible to use overlapping locals, which occasionally helps.
 - Use "*volume maximum*" in CHIMERAX to combine with best local for each subdomain.
 - Use default/equal scale for all maps because density balance is not important at this stage.
5. Run Real Space Refinement in PHENIX against manual composite
 - Check for alignment of all 2° elements
 - Rebuild as needed to get backbone to match everywhere. Sidechains not important at this state.
6. Use that preliminary refinement to generate a composite map in PHENIX using the same maps used to make the manual composite.
7. Perform *aredecon* run on the resulting PHENIX composite map.
 - Initially run *aredecon* using default parameters.
 - Then run parameter search to select best smoothing and non-linearity parameters.
 - Rerun *aredecon* using optimized parameters.
 - Run inline PHENIX sharpening during that aredecon run.
 - Run external model-based PHENIX sharpening on the output map.
 - Compare map quality in key regions to select best map for presentation or seeding the next iteration of local refinement.

Composite map generation:

- If there are prior high-quality locals for a similar conformation, try those restraint parameters initially (typically somewhere between 1/1 and 5/5).
- Restraint parameters usually need to be manually searched by iterating values up/down, repeating the local refinement, and manually inspecting those maps.
- Align all locals to the consensus map from which they were derived. (If there are sub-locals, align those to the locals from which they were derived.)
- Resample all local maps on the grid of the consensus map, and save those maps to disk for using in composite map generation in PHENIX.
- Adjust level of consensus map so that there is a very small amount of noise visible in the solvent region, and then manually scale all locals to a similar density level (by eye).
- Repeat alignment of all locals to the consensus map after adjusting all the levels to match.
- Calculate relative scale factors for each local map by dividing its level value at the chosen contour level by that of the consensus map.
- Run volume maximum on the consensus and all locals using those scale factors, *e.g.*, “volume maximum #1-7 scaleFactors 1.0, 0.71,0.71,0.79,0.43,0.42,0.54”.
- Load the resulting manual composite map into COOT and align all subdomains to the density as well as any protein segments displaced from it.
- Refine the resulting model against the manual composite map in PHENIX.
- Use that output together with the aligned consensus and locals to calculate a composite map in PHENIX.
- Complete refinement of the model against the composite map in PHENIX, possibly after running it through DeepEMhancer.

Generation of local resolution map from a composite map:

- Generate half-map composites using the map_half_A and map_half_B outputs from the same consensus and local refinement jobs used for the full composite.
- Align each local half-map to the corresponding consensus half-map (*i.e.*, A locals to A consensus and B locals to B consensus).
- Make composite maps in PHENIX separately for the half_A set and half_B set using the same PDB file used to generate the overall composite map.
- Import the composite half-maps into cS making sure to select either “half_map_A” or “half_map_B” under “Type of volume being imported.”
- Import the overall composite in cS as usual.
- Run a Local Resolution job in cS:
 - Initially load the imported overall composite map as input by dragging/dropping.
 - Then use the drop-down tab on the import sub-menu to replace the half-map inputs with the half-map volumes shown in the output tabs in the import jobs.
- Load the locres output map from that job into ChimeraX together with the overall composite map to color-code the local resolution on the surface of the composite:
 - Choose an appropriate contour level, matching density in the composite map region where locals were not performed to that in the parental consensus map if comparison is needed.
 - Hide dust if needed (typical threshold = 6).
 - Go to the “Volume” -> “Surface color” menu, activate the “Options” button, and select 5 colors and rainbow palette.
 - Enter appropriate resolution values for each color (*e.g.*, 2.6, 3.2, 3.8, 4.4, 5.0)
 - Select Color surface by “volume data value”, and select the corresponding locres map in the new dropdown menu that appears.
 - Hit “Color”.
 - Align view and save it if comparison to other maps is needed.
 - To output image for publication, select “Presets” -> “Publication 1 (Silhouettes)”, then select “Soft” lighting under the “Graphics” tab.

Miscellaneous cryoSPARC installation/maintenance notes:

- **Set an ssd reserve amount that will not be occupied by the cryoSPARC cache (in MB):**

```
> sudo /data/emsoft/cryosparcV2/cryosparc2_worker/bin/cryosparcw connect --master yellow.biology.columbia.edu --worker yellow.biology.columbia.edu --port 39200 --update  
--ssdpath /ssd/cryosparc_temp --ssdreserve 100000
```

Instructions to process a composite map using *ardecon*:

1. Prepare the input files needed to run *ardecon*:

- Composite full map from *PHENIX*.
 - Align local maps to the consensus map, and resample them on the grid of the consensus map before saving to disk.
 - Generate a crude manual composite from the aligned consensus and local maps using “volume maximum” in ChimeraX.
 - Make sure the PDB model is well-aligned to the manual composite before running PHENIX composite map generation routine.
 - May be best to run Real Space Refine in *PHENIX* to ensure good alignment.
 - Run composite map generation in *PHENIX*.
- Composite half maps from *PHENIX*.
 - Align all half-maps, both the consensus and all of the locals, to the full composite map from PHENIX, and resample them on that grid before saving to disk.
 - Run *PHENIX* separately on the aligned half-A and half-B maps to generate composite half maps.
 - Use the same PDB file used to generate the composite full map.
 - This procedure is the same one used to generate half maps for local resolution calculation.
- Generate a Molmap at 3.5 Å in *ChimeraX* from the PDB model used to generate the composite maps.
- Use *cryoSPARC* to generate a mask from that Molmap using 2 Å dilation and 12 pixel padding.
- Upload the full and half (A & B) composite maps and the Molmap and mask file to the workstation running *ardecon* (5 files total).
 - Open a terminal window, and cd to the directory with those files on your local computer.
 - `>sftp uid@east.biology.columbia.edu`
 - `cd` to the directory in which you want to run *ardecon*.
 - `>put *.mrc`

2. Do an initial *ardecon* run following instructions at <https://github.com/yifancheng-ucsf/AR-Decon>:

- `>conda activate ardecon`
- `> source /usr/local/phenix-dev-3092/phenix_env.sh`
- `>ardecon AltNBD1_RSR149_composite_half_A.mrc AltNBD1_RSR149_composite-map_half_B.mrc AltNBD1_RSR149_composite-map.mrc --mask AltNBD1_RSR_Ordered_04232025a-coot-4_real_space_refined_149_J3269-AlignedMask.mrc --post phenix`
- The program will create an *ardecon* subdirectory containing all results.

3. Once the initial *ardecon* run is finished, optimize deconvolution parameters following instructions at <https://github.com/yifancheng-ucsf/AR-Decon>:

- `>param_search AltNBD1_RSR149_composite-map.mrc AltNBD1_RSR_Ordered_04232025a-coot-4_real_space_refined_149_J3269-AlignedMolmap.mrc ardecon/AltNBD1_RSR149_composite-map_HalfMapdFSC3d.otf --mask AltNBD1_RSR_Ordered_04232025a-coot-4_real_space_refined_149_J3269-AlignedMask.mrc`
- The program will create a "ParamSearch" subdirectory containing all results.
- The gallery of dFSC plots produced by the `param_search` routine needs to be examined to identify the parameters giving the best FSC.

4. Re-run *ardecon* using optimized deconvolution parameters:

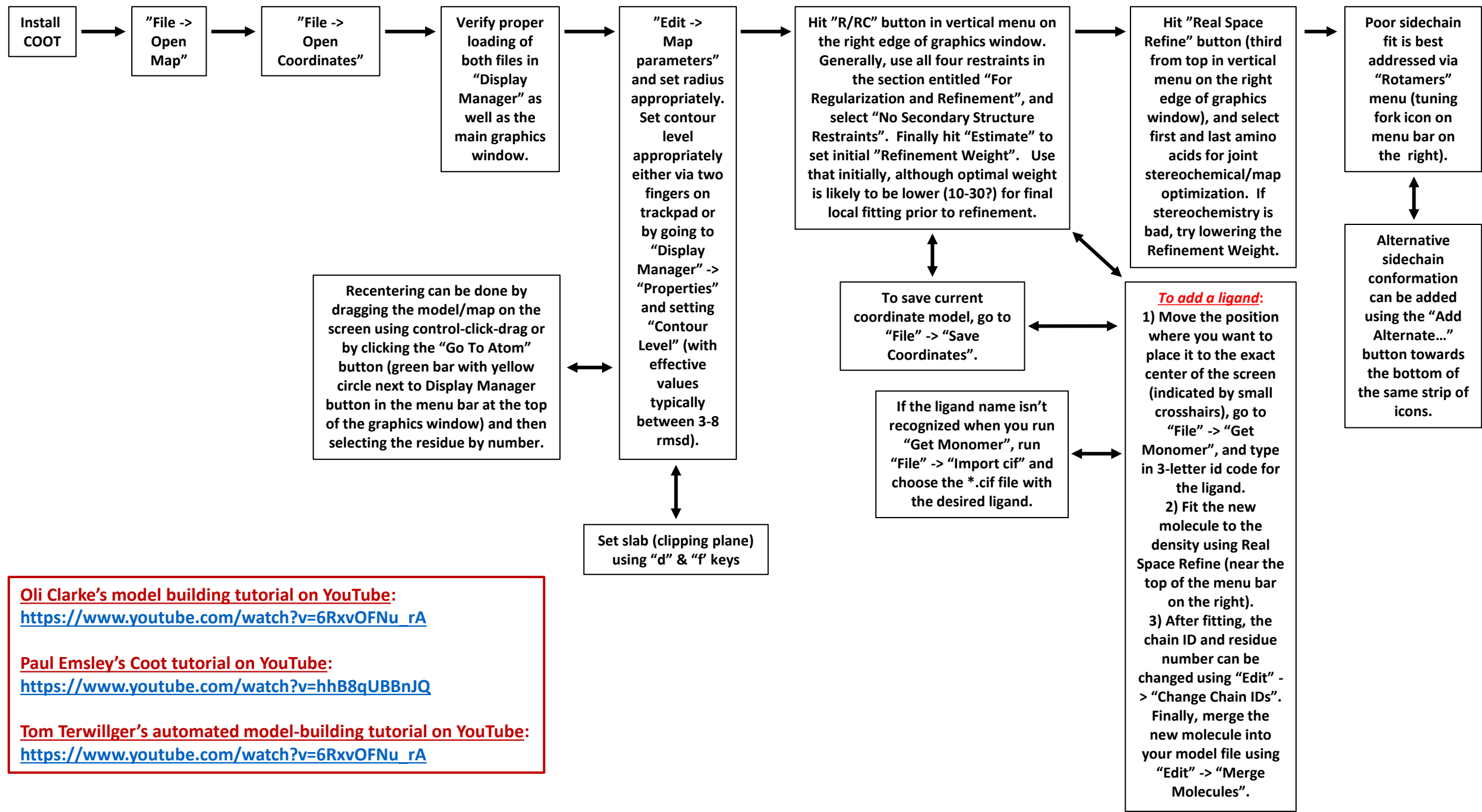
- `>cd ParamSearch`
- `>ardecon AltNBD1_RSR149_composite_half_A.mrc AltNBD1_RSR149_composite-map_half_B.mrc AltNBD1_RSR149_composite-map.mrc --mask AltNBD1_RSR_Ordered_04232025a-coot-4_real_space_refined_149_J3269-AlignedMask.mrc --smooth ??a --nonlin ??b --post phenix &`
- The final “&” runs the job in the background. Get the job number by executing the `jobs` command (`>jobs`), and then disown the job (`>disown %job#`) to prevent termination upon logout.

5. Transfer the new *ardecon* map to local computer for model-based sharpening in PHENIX.

6. Import that PHENIX-sharpened *ardecon* map into *cryoSPARC*, and use it as a reference volume for Non-Uniform (NU) Refinement of the particles used to generate the input maps.

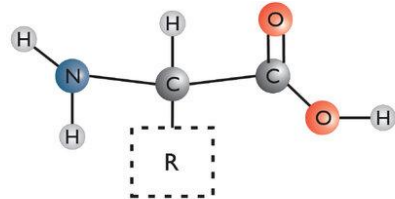
- Use 4 Å initial resolution for NU.
- Run NU refinement ± 8 extra final passes. The map without extra final passes is likely to be best in mobile (sub) domains, while the map with them is likely to be best in the well-ordered core.

Manual model building in COOT:



20 canonical amino acids:

Hughes & Dougan (2016) *Reports on Progress in Physics* 79:7



5 canonical nucleotides:

Nonpolar			Polar	
Glycine (G)	Alanine (A)	Proline (P)	Serine (S)	Threonine (T)
Valine (V)	Leucine (L)	Isoleucine (I)	Glutamine (Q)	Asparagine (N)
Sulfur containing		Aromatic		
Cysteine (C)	Methionine (M)	Phenylalanine (F)	Tyrosine (Y)	Tryptophan (W)
<i>Somewhat polar</i>	<i>Apolar</i>			
Positively charged*		Negatively charged*		
Histidine (H)	Lysine (L)	Arginine (R)	Aspartic acid (D)	Glutamic acid (E)

Coordinate refinement protocol/tips:

- If there are prior high-quality refinements for some or all regions of the structure/map, use those to create a starting model for computational refinement:
 - Identify the regions (domains/subdomains/helices/sheets) in the protein that move as rigid bodies and split those into separate PDB files.
 - *Align prior model to one region of new map and determine regions in the map where the old model does not align with the new map.*
 - *Note the exact residue numbers at the boundaries where the old model doesn't because those sites will need to be handled differently.*
 - *Use PyMOL or a text editor to extract the matching region into a new PDB file (excluding any residues that don't match).*
 - *Be sure to retain at this stage any rigidly bound ligands present in the target map and to exclude those that are not.*
 - *Repeat for all regions in which the old model locally aligns to the new map.*
 - Align each resulting PDB file with the new map, and write out the aligned coordinates.
 - If those PDB files have anisotropic B-factor records, delete those lines before grafting. (UNIX command “grep -v ANISOU current.pdb > updated.pdb”)
 - Also safest to delete all SEQRES, HELIX, STRAND, and CONNECT lines from those PDB files.
 - Graft the coordinates from all of the aligned domains/subdomains/2°-structure elements into a single model using a text editor.
 - Make sure that no residues are duplicated and that no residues clash at the junctions between the regions in the different constituent PDB files.
 - If there are clashing residues, retain those closest to the density in the new map, and delete the others.
 - If residues are missing at the junctions, best to add them in now from whatever model matches best in that region (if not, must be done later in COOT).
 - Read the grafted coordinate file into PyMOL, sanity-check it visually, and write it out to clean-up the atom numbering.
 - Check that the resulting composite model properly matches all regions of the new map. (Best to reload that map in ChimeraX to rule-out origin shift).
- Set up any needed restraints for PHENIX refinement:
 - Put coordinates for each rigid domains/subdomains/helices/sheets to be restrained into a separate PDB file.
 - Make sure those PDB files do not contain any residues that should not be forced to match the reference conformation closely.
- Prep the PDB file containing the grafted/clean coordinates for refinement by running it through READYSET in PHENIX:
 - Be sure to select “Add hydrogens to model if absent”, “Optimize final geometry of hydrogens”, and “Generate ligand restraints”.
 - Otherwise use defaults.
 - Include as inputs a cif directory or individual cif files with geometric restraints if there are any non-standard ligands in the model.
- Run RealSpace Refinement in PHENIX:
 - Input map file and all the PDB and cif files generated by ReadySet, including any cif files generated for non-standard ligands.
 - Input any reference model files to be used for restraints, and make sure each of those is designated as a “Reference model” in the input GUI.
 - Under Refinement Settings, set “Rotamers...” menu to Fit=“outliers_or_poormap”, “Enabled”, Sigma=3.0, Target=“fixoutliers”, & Tuneup=“outliers_and_poormap”.
 - Under Refinement Settings -> “All parameters...” -> “Search parameters...”, search “nonbonded” and set nonbonded weight to 750.0.
 - Add any occupancy refinement groups (e.g., “chain c and resseq 1095:1120”) under Refinement Settings -> “All parameters...” -> “Atom selections” -> “Occupancy refinement” text box.
 - Add non-standard geometry constraints (e.g., peptide bond at an internal deletion site) using proper PHENIX syntax in a “.phil” input file.
 - Otherwise use defaults.

Streamlined transfer of a well-refined model to fit a new map in a different conformation:

- Rigid-body refinement and then local real-space refinement of individual subdomains and secondary structural elements:
 - Reasonable segments in the transmembrane domains of hCFTR are residues 1-42 (LassoNter), 42-77 (LassoCter), 77-218 (TMHs 1-3), 218-333 (TMHs 4-5), 333-376 (TMH 6), 376-388 (TMD1-NBD1 linker), 844-886 (TM7), 918-1012 (TMHs 8-9), 1012-1125 (TM10-11), & 1125-1173 (TM12).
 - Reasonable segments in NBD1 of hCFTR are residues 388-493 + MgATP (1st segment of CORE1), 493-565 (AHD1), 565-579 (2nd segment of CORE1), 579-597 (Walker B1), 597-638 (3rd segment of CORE1).
 - Reasonable segments in NBD2 of hCFTR are residues 1202-1291 + Na/MgATP (1st segment of CORE2), 1291-1364 (AHD2), 1364-1376 (2nd segment of CORE2), 1376-1394 (Walker B2), 1394-1443 (3rd segment of CORE2).
- Perform local real-space refinements to repair possible backbone distortion across the boundaries between all of the segments that were separately refined:
 - For hCFTR, residues 37-47, 72-82, 213-223, 328-338, 371-381, 383-393, 488-498, 560-570, 574-584, 592-602, 1007-1017, 1020-1030, 1286-1296, 1359-1369, 1371-1381, & 1389-1399
- Check for and try to fix any Ramachandran violations that arise in the junction segments.
- Run PHENIX real-space refine using 750 non-bonded weight and standard rotamer refinement parameters.

Finalizing coordinate models for PDB deposition:

- Check every residue for alignment of its C α carbon and sidechain with the density.
 - Run real space refinement in COOT including ± 2 or ± 3 residues if the alignment is off, and use the "rubber band" feature to center both backbone and sidechain.
 - The C α carbon should be close to the center of the tubular density following the backbone in medium-resolution maps.
 - Geometric distortions in the model most frequently arise from either $\Delta\Phi/\Psi$ backbone peptide flips or having the wrong rotamer.
 - Computational refinement of the wrong rotamer will tend to pull it into the density by distorting the backbone geometry.
 - Therefore, after refinement, the correct rotamer can be skewed from the density due to the backbone distortion.
 - This effect means it can be worth performing real-space refinement a few different rotamers to see which give the best fit after local computational optimization.
- Check for $\Delta\Phi/\Psi$ variations *versus* a high-resolution reference model or models if available, and match backbone geometry to the reference model if consistent with density.
 - The program phi_psi.py written by Polina Goldberg is a convenient way to do this.
 - A large ($\geq 120^\circ$) $\Delta\Psi$ at one residue followed by a large $\Delta\phi$ at the next, generally means the peptide bond between them should be flipped.
 - Large $\Delta\Phi/\Psi$ values 2-4 residues apart often mean two non-adjacent residue need backbone peptide flips.
 - Positive $\Delta\Phi/\Psi$ values are relatively rare for all residues other than GLY and ASN and only common for GLY.
- Procedure to finalize the lipid model:
 - Use PyMOL to align the protein in the VShaped model above to the model to be completed. Save the aligned model, and open that in COOT.
 - For lipids in the model being finalized that align reasonably with lipids in the VShaped model, change their Chain ID and residue number to match those in the VShaped model.
 - If different species are aligned with one another, convert to the species in the VShaped model if that's as good a match or a better match to the density for the model being finalized. If the aligned species from the VShaped model is a worse fit to the density, don't change its identity but still change the Chain ID and residue number to the species at that site in the VShaped model.
 - Lipid species that are present in the model being finalized that have no match in the VShaped model should be given residue numbers after the last number in the VShaped model that has the same Chain ID.
 - Check for any additional unmodeled blobs. If any are found, model them, and assign them residue numbers after the last number in the VShaped model that has the same Chain ID.

Organizing refinement data for EMDB/PDB deposition:

- Collect the map used for refinement, corresponding half-maps, and consensus map and all local maps plus their corresponding half-maps if refining against a composite map.
- Get a screenshot of final PHENIX real space refinement summary to have a record of the d99 and model/map FSCs, and save it in the same directory as the maps.
- Run the PDB validation server (<https://validate-rcsb-1.wwpdb.org/>), and save its report in the same directory as the maps.
- Prepare any needed REMARK file.

PDB file manipulation widgets:

Remove anisotropic B-factor records from a PDB file:

```
grep -v ANISOU input.pdb > output.pdb
```

Report on multiple conformations in a PDB file:

```
awk '/^ATOM/{if (substr($0,17,1) != " ") print}' input.pdb
```

GLOBUS crib notes:

Program location (potential a soft link to installation):

/usr/etc/globus/globusconnectpersonal-3.2.0/ #Yellow???

/mnt/data4/globusconnectpersonal-3.2.0/globusconnectpersonal #West

Otherwise find globuspersonalconnect:

>locate globuspersonalconnect

Globus website URL:

<https://app.globus.org/>

Hunt lab share-points/collections at CUMC:

Krios1_Hunt

Krios2_Hunt

Verify no Globus endpoint is currently running (but note that path to Globus Personal Connect varies on different workstations):

/usr/etc/globus/globusconnectpersonal-3.2.0/globusconnectpersonal -status

Verify that Globus personal connect has read access to the needed data directory, which is specified by the file ~/.globusonline/Ita/config-paths :

-- Modify the allowed directory list if needed

-- Format of config-paths file is as follows:

~/,0,1

/mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR,1,1

-- Active Globus endpoint on the Hunt lab West workstation as of July 16, 2025:

If not, run Globus setup:

/usr/etc/globus/globusconnectpersonal-3.2.0/globusconnectpersonal -setup

-- Enter URL returned by setup into a new tab in a browser connected to Globus.

-- Then give the endpoint clear name, e.g., "HuntLab-schrogu1".

-- Then enter the authcode given by the Globus browser on the workstation.

-- Then enter on the server the same endpoint name given in Globus.

-- Activate endpoint most recently set up:

/usr/etc/globus/globusconnectpersonal-3.2.0/globusconnectpersonal -start >GlobusLOG 2>&1

-- Send job to background with ^Z.

-- Run job in background:

bg %job#

-- Disown the job:

disown %job#

-- Make destination directory under an allowed directory on the server.

Return to Globus webpage to setup transfer:

-- Enter source collection sharepoint and navigate to target folder.

-- Active "Transfer or Sync".

-- Search for target collection, e.g., "HuntLab-schrogu1".

-- Navigate to target directory on the server.

-- Set up timer to desired frequency/number of repeats, and initiate transfer.

Protocol to setup shared Endpoint with access to specific local directories???

Schrödinger Suite job launch & monitoring commands and output analysis widgets:

The following lines must be added to your ~/.bashrc and ~/.profile files before running Schrödinger Suite software with the path variables edited to match that workstation:

```
export PATH="$PATH:/opt/schrodinger2025-3/"
export SCHRODINGER="/opt/schrodinger2025-3"
export SCHRODINGER_SCRATCH="/ssd/jfh21/"
```

Launch a Glide screen on one of our LINUX workstations:

```
>"${SCHRODINGER}/glide" glide-dock_name.in -OVERWRITE -adjust -HOST localhost:56 -TMPLAUNCHDIR >LOGdate 2>&1 &
>jobs
>ls -lath
>ps -ef | grep -i schr
>${SCHRODINGER}/jobcontrol -list
>htop
```

Check status of ongoing GLIDE run:

```
>ls /scratch_disk/schrodinger_scratch/uid/job_name/*[0-9].csv | wc
> cat /scratch_disk/schrodinger_scratch/uid/job_name/*[0-9].csv | wc
> cat /scratch_disk/schrodinger_scratch/uid/job_name/*[0-9].csv | awk 'BEGIN{FS=","}{if ($9<-8.0) print}' > GlideScoresUnder8.csv
```

Count lines in GLIDE output csv file with Glide Scores under -8.0:

```
awk 'BEGIN{FS=","}{if ($9 < -8.0) print}' report.csv | sort -t ',' -k 9n | wc
```

Count lines in final IFD output csv file with Glide Scores under -12.0:

```
awk 'BEGIN{FS=","}{if ($4 < -12.0) print}' report.csv | sort -t ',' -k 4n | wc
```

Useful chemical screening software and websites:

Smiles converters:

<https://www.leskoff.com/s01810-0>

General cheminformatics software:

<https://www.rdkit.org/>

Setting up Schrödinger IFD-MD:

- Make sure the user has a scratch directory with world read/write permission on an appropriate scratch disk, typically a high capacity ssd:
 - “df -h <return>” to locate ssd
 - “ls -tlah /ssd/uid <return>” to check the directories on the ssd and their permissions.
 - If needed create the scratch directory and set its permissions, and make sure there is a directory matching your account name and that is world readable and writable:
 - “sudo mkdir /ssd/uid && chmod -R 777 /ssd/uid <return>” with /ssd/uid being path to the scratch directory.
- Define the SCHRODINGER and SCHRODINGER_SCRATCH variables in both interactive and non-interactive shells, and also add the SCHRODINGER directory to your PATH.
 - Use the nano editor to add the following three lines to both your ~/.profile and ~/.bashrc files (with the directory string set to the current Schrödinger suite installation):

```
export PATH="$PATH:/opt/schrodinger2025-3/"
export SCHRODINGER="/opt/schrodinger2025-3"
export SCHRODINGER_SCRATCH="/ssd/uid/"
```
- Verify that the SCHRODINGER variables are set and that your PATH contains the SCHRODINGER – logout and log back in if it isn’t:
 - “echo \$PATH <return>”
 - “set | grep SCHR <return>”
- Set up a directory with world read/write access on a large data drive in which to run IFD-MD (*e.g.*, /data/hCFTR/ChemicalScreening/IFD_MD/IFD_MD_MonthYear/).
- Move command “.sh” file and Maestro input files to that directory using sftp.
- Edit the command file to set n_cpu to “56” and n_gpu to “4” (for jobs on hudson.biology.columbia.edu) and gpu_subhost to “localhost”.
- Print the .sh file to the screen using the cat command, copy the full content of that file, and paste it onto the command line followed by “>LOG 2>&1 & <return>”.
- Get the job number by entering “jobs <return>”.
- Disown the job by entering “disown %1 <return>” (assuming the IFD-MD jobs is #1).
- Check whether any Schrodinger processes are running by entering “ps -ef | grep -i schr <return>”.
- Check cpu usage using “htop <return>” (q to exit that screen) and “nvidia-smi <return>” to check GPU usage.
- Enter “cat LOG <return>” to check stdout and stderr from the ifd-md command.
- Enter “ls -talh <return>” to check for presence of a .log file.
- Enter “ls -talh /ssd/uid <return>” to check that files are being written to the scratch directory.
- Enter “more schrodinger_job_file.log <return>” to check the job progress and status (assuming schrodinger_job_file.log is the log file name for the current IFD-MD run).
- The progress of the job can also be monitored using “tail -f schrodinger_job_file.log <return>” (“^c <return>” to exit).
- Check for errors in the log file using “grep -l err schrodinger_job_file.log <return>”.

Standard structural biology software package installations:

Miniconda (API that facilitates accessing/running Python libraries):

<https://www.anaconda.com/download/success>

BEdit (convenient text editor including basic code formatting):

<https://www.barebones.com/products/bbedit/download.html>

PyMOL:

<https://pymol.org/>

ChimeraX:

<https://www.cgl.ucsf.edu/chimerax/download.html>

PHENIX:

<https://phenix-online.org/download>

XQuartz:

<https://www.xquartz.org/>

COOT via CCP4:

"Install appropriate version (see below) of XQuartz first!"

<https://www.ccp4.ac.uk/download/#os=macos>

phi_psi.py program to analyze backbone geometry shifts:

<https://github.com/huntmolecularbiophysicslab/CryoEM>

Software tutorial screencasts:

PyMOL:

https://www.youtube.com/watch?v=Zl0o_Qi4s08&list=PL3QqtTa0J274K3k3YxypuUifJCPuLW1_y&index=9&pp=gAQBiAQB

Protein structure screencasts and slidedecks:

Ramachandran angles:

https://www.youtube.com/watch?v=bdtZntuf1aY&list=PL3QqtTa0J274K3k3YxypuUifJCPuLW1_y&index=19&pp=gAQBiAQB

Beta structures:

https://www.youtube.com/watch?v=IY2-MXLmQgQ&list=PL3QqtTa0J274K3k3YxypuUifJCPuLW1_y&index=8&pp=gAQBiAQB

Analysis of domain rotations:

https://www.youtube.com/watch?v=rn7ALW4IDIM&list=PL3QqtTa0J274K3k3YxypuUifJCPuLW1_y&index=3&pp=gAQBiAQB0gcJCbAJAYcqIYzv

John's lecture notes for BIOL GU4323 *Biophysical Chemistry I*:

<https://www.dropbox.com/scl/fi/f5ytoxx8hiicfnb6ckyus/HuntBCHMGU4323Thermo09032024a.pdf?rlkey=kggmzcl4gkhyzqhcgjni601ke&dl=0>

UNIX crib notes:

cryoSPARC server addresses:

<http://yellow.biology.columbia.edu:39200/> [sgpu]
<http://ganges.biology.columbia.edu:39000/>
<http://east.biology.columbia.edu:39000/>
<http://west.biology.columbia.edu:39000/>

Disk farm computer:

volga.biology.columbia.edu

Hunt lab Synology drives:

<http://128.59.235.5:5000/>
<http://128.59.243.40:5000/>

Reference volume/map directory on yellow:

/home/jfh21/hCFTR

- Procedure to prevent a backup job running in a terminal session from crashing upon logout:
 - Redirect stderr to stdout = 2>&1
 - Redirect both outputs to a log file = >LOG
 - Run the job in the background = &
 - Disown the job = disown %job#
 - Example (assuming job# is 1):
 - \$ rsync -avh source_directory target_directory 2>&1 > LOG & <return>
 - \$ more LOG <return>
 - \$ jobs <return>
 - \$ disown %1 jobs <return>
 - \$ ps -ef | grep rsync <return>

- If a cryoSPARC job crashes, log into the workstation, run 'df -h' to check disk mounting and then 'du -h', and check disk usage on /scratch or /ssd and whatever disk holds the working directory in cryoSPARC.

UNIX command for remote secure shell login (uid = username):

\$ ssh uid@yellow.biology.columbia.edu

UNIX command for remote file transfer (uid = username):

\$ sftp uid@yellow.biology.columbia.edu

UNIX command to change password:

\$ passwd

UNIX command to check mounting & available disk space on system drives:

\$ df -h

UNIX command to get concise directory listing:

\$ ls

UNIX command to get more complete and verbose directory listing:

\$ ls -laht

UNIX command to get summary of total disk usage of directory tree:

\$ du -hs <directory_name>

UNIX command to print working directory name:

\$ pwd

UNIX command to change directory :

\$ cd <directory_name>

UNIX command to make new directory :

\$ mkdir <directory_name>

UNIX command to check for mounted devices (disks):

\$ lsblk

UNIX help listing (print manual page to screen):

\$ man command

UNIX command to transfer data tree between servers/directories:

\$ rsync -avrhP source_dir uid@server.biology.columbia.edu/target_dir

UNIX/NVIDIA command to show GPU status:

\$ nvidia-smi

UNIX command to create a data archive file including original of softlinks:

\$ tar -cvhf tar_archive_filename.tar target_directory_name

UNIX command to create a “softlink” to an existing file or directory:

\$ ln -s source_file_or_directory softlink_file_or_directory_name